



**Faculty of Mechanical
Engineering and Robotics**

**Department of Robotics and
Mechatronics**



Signal processing and identification in control of mechatronic devices

Topic: Image segmentation and shape analysis

Goal: Introduction of students to image segmentation methods and description of geometric shapes by image features. Image segmentation methods based on thresholding and texture analysis are discussed.

Problems:

1. Intensity level based segmentation (thresholding),
2. Description of geometric shapes based on image features available in MATLAB programming environment (function *regionprops()*),
3. An application of image features for shape detection and recognition,
4. Image segmentation methods based on texture and texture filtering,

1. Object recognition on binary images

Object recognition is one of the most important tasks of computer vision. In a lab, shape recognition on binary image will be presented. In order to analyze and detect object, the image has to be prepared first. Noise has to be removed and grayscale to binary conversion must be performed. Objects are represented by pixels of value 1 and background as pixels of value 0. First, all objects are labeled using **bwlabel()** function in MATLAB.

```
[im_lb, num] = bwlabel(im, nb);
```

Function takes binary image and type of neighborhood of object pixels. 4- and 8- pixel neighborhoods can be used. Next, the geometric parameters describing shapes are computed using **regionprops()** function. The syntax of the function is as follows:

```
features = regionprops(im_lb, properties);
```

where the variable *properties* is a string of characters describing which geometrical features we want to obtain. The simplest choice of properties are 'all' and 'basic'. The first one means that all features has to be calculated and the second one gives only the basic one, according to the authors of the function. The *features* variable is a matrix of MATLAB structures which stores the numerical values of all features computed for all objects on the binary image.

Problems 1.

1. Chose parameters of image filters so all noise is removed from the image.
2. Test regionprops() function on images with objects extracted. Test all geometrical features, what they mean and how their values are change when you change the analyzed shapes.
3. Using the knowledge on geometrical features, chose the smallest set of features which is necessary to detect an object on the image given by the teaching assistant.
4. Use object features for classification of objects present on the image into categories.

2. Texture-based segmentation

Threshold-based segmentation works when both objects and background is more or less homogeneous in intensity value. When the objects are textures, the method is very hard to practical use and may generate erroneously segmented results. For segmentation of images based on texture, MATLAB provides set of three filters: **entropyfilt()**, **stdfilt()**, **rangefilt()**. Each of the filters computes for each pixel of the image, a value which is a function of pixels

neighboring with the analyzed one. Default neighborhood has a size 3x3, but the size can be chosen by an user. As an example, **stdfilt()** filter computes standard deviation of pixels belonging to a neighborhood of analyzed pixel. It can be applied to extract regions which are homogeneous and the ones which are characterized by large variation of intensity. Another popular texture measure is Grey-level co-occurrence matrix (GLCCM). In MATLAB, it can be computed by a function **graycomatrix()**. Computed matrix does not give information on the statistics of pixels in the image. Coefficients based on the matrix has to be calculated. MATLAB has function **graycoprops()** which computes four different coefficient which describe pixels of a textures regions.

Problems 2.

1. Test texture filters for different pixel neighborhoods. What are the differences in computation results?
2. Perform segmentation of images based on texture analysis.
3. What are the parameters used for computing from GLCC matrix? What are the physical meaning of the parameters? Test all 4 coefficients computed for different types of textures. Use them for image segmentation.